
Training data
$\{(x^{(s)}, y^{(s)})\}_{s=1}^{N}$
Parameterized model
$\widehat{y} = f_\theta(x)$
Prediction
$\widehat{y}$
Loss function
$\ell(\widehat{y}, y)$
Backpropagation
$\nabla_\theta \mathcal{L}$
Parameter update
$\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}$